COMPRHENSIVE ANALYSIS OF WATER QUALITY OF RAPTI RIVER IN GORAKHPUR REGION

Devendra Pratap Singh[1], Krishna Jaiswal[2], Suraj Srivastav[3], Ajeet Yadav[4], Mr. Rajan Shukla[5]

[1,2,3,4]*Under Graduate Student, Civil Engineering, Buddha Institute of Technology, CL1, Sector-7, GIDA, Gorakhpur, Uttar Pradesh, 273209*

[5]*Assistant Professor, Department of Civil Engineering, Buddha Institute of Technology, CL1, Sector-7, GIDA, Gorakhpur, Uttar Pradesh, 273209*

ABSTRACT: The Rapti River is a vital freshwater resource for irrigation, drinking, and maintaining ecological balance in the region. However, the river is facing increasing pollution due to anthropogenic activities, industrial effluents, and agricultural runoff. This study evaluates the physio-chemical parameters of the river water to assess its quality. Key parameters, such as pH, turbidity, hardness, alkalinity, and chloride content, were measured at various locations along the river. We recommend regular monitoring, effective wastewater management, conservation of natural habitats, and sustainable agricultural practices to mitigate pollution and ensure the river's water is safe for drinking, irrigation, and other uses. This study highlights the need for integrated water resource management to protect the Rapti River's water quality and promote sustainable development.

Keywords: Rapti River, water quality of Gorakhpur, pollution, environmental assessment, water parameters, water management, river pollution.

1. INRODUCTION

The Rapti River plays a crucial role in the Gorakhpur district, providing essential water resources for agriculture, industry, and domestic purposes. However, the river is under serious pollution threats caused by various human activities. The pollution of the Rapti River has profound environmental and public health consequences, impacting both aquatic ecosystems and the human populations relying on it. Originating from the lower Himalayas and flowing into the Gangetic plains, the river sustains a wide range of biological systems and serves as a primary water source for surrounding communities. Increasing human activities, such as urbanization, agriculture, and industrial development, have led to a decline in water quality, necessitating a comprehensive assessment.

Evaluating the water quality is crucial to determine its suitability for drinking, irrigation, and industrial use. This study aims to analyse selected water quality parameters at key locations along the Rapti River to assess the spatial variation in water quality and pinpoint sources of pollution. Water quality parameters such as Ph, dissolved oxygen (DO), turbidity, temperature, biochemical oxygen demand (BOD), chemical oxygen demand (COD), nitrates, phosphates, and microbial content provide insights into the health of aquatic systems and the presence of pollutants. These parameters help identify contamination sources, track changes over time, and guide water management policies.

2. LITERATURE REVIEW

2.1 Sahni, P. & Varshney, D.[1] (2024)- Investigated that river water has been used as a source of drinking water. River water is now a source of hydropower generation, irrigation, aquaculture, navigation, transportation, and supports socio-economic activity, human settlement and aquatic living forms. Due to increase in industrialization, urbanization, mining, sewage disposal and rampant use of technology, Yamuna River water is becoming polluted day by day. This may possess a health risk to several rural communities as well as urban localities who rely on the river water primarily as their source of domestic water and irrigation. The present study showed an urgent need for a continuous pollution monitoring and treatment programmes of Yamuna River water in India.

2.2 Alam, JB et al.[2] (2007)- Water samples have been collected from a part of Surma River along different points and analyzed for various water quality parameters during dry and monsoon periods. The study was conducted within the Chattak to Sunamganj portion of Surma River, which is significant due to the presence of two major industries-a paper mill and a cement factory. The results from data analysis show that, the water is certainly unfit for drinking purposes without any form of treatment, but for various other surface water usage purposes, it still could be considered quite acceptable.

2.3 Yadav, D. & Pandey, G.[3] (2021)- Gorakhpur is situated from bank of Rapti River. Rapti River is joined at Sohgaura with Ami River which is highly polluted as compare to Rapti River. Rapti River has a confluence into Ghaghra River at Kaparwar Ghat. The present study is conducted to determine the water quality of Rapti River using physical and Chemical parameters including pH, Turbidity, Total dissolved solids, Dissolved Oxygen (DO), and Total Hardness. The study also reveals that the water quality of Rapti River is highly deteriorated due to anthropogenic activities such as urbanization, construction activities, agricultural activities, discharge of untreated sewage and disposal of solid wastes into river.

2.4 Rai, A.[4] (2015)- Ami River is highly polluted as compare to Rapti river. The impact of water of river Ami on ground water quality of river Rapti abstracted through shallow and deep hand pumps placed in close vicinity of river Ami and river Rapti. There were three stations selected for the study i.e. Malav. Sohagaura and Kauriram. In this study, the water quality parameters of physical, chemical and biological characteristics of this ground water quality parameters from shallow depth and pumps and Indian Mark- hand pumps were studies in Kauriram (before meeting Rapti River), Malav (before meeting Ami River) and Sohagaura (after meeting Ami River and Rapti River).

2.5 Pandit, P. & Fulekar, M.H.[5] (2017)-Investigated that statistical analysis such as principal component analysis was employed to evaluate the status of water quality for nine monitoring regions. The present study was carried out to determine the physicochemical parameters of water of different coastal region of Gujarat India during pre-monsoon time. Results provide baseline information on the environmental conditions and anthropogenic impact along the coastal region of Gujarat. Thus, the present study illustrated the usefulness of principal component analysis a statistical tool for the analysis and interpretation of complex data set and water assessment.

3. METHODOLOGY

3.1. Sample collection: We have collected water sample from four places which are present in GIDA, Gorakhpur, for testing their properties.

- Rajghat
- Barhuan
- Mokshdham
- Muhamdpur maphi

Figure 3.1Sample collection

3.2. Selected Parameters: The water samples were analyzed for various physico-chemical parameters, including:

- pH
- Alkalinity
- Hardness
- Turbidity
- Acidity
- Total Dissolved Solids

3.3. Parameters Analyzed:

a. pH:

- pH is a measure of the acidity or basicity of a solution. In the context of water quality, pH is an important parameter to measure.
- In water quality monitoring, pH is measured to:
- Determine the acidity or basicity of the water
- Assess the potential for corrosion or scaling
- The acceptable pH range for drinking water is typically between 6.5 and 8.5.

b. Alkalinity:

Alkalinity is the ability of water to resist changes in pH when acids or bases are added. It's a measure of the concentration of bases, such as bicarbonate (HCO3-), carbonate (CO32-), and hydroxide (OH-), that can neutralize hydrogen ions (H+). • Alkalinity can be measured using various methods, including:

1. Titration: This is a laboratory-based method that involves adding a strong acid to a water sample until the pH reaches a predetermined endpoint.
2. Colorimetric methods: These methods use a color-changing reagent to measure alkalinity.
3. Ion chromatography: This method uses a chromatographic technique to separate and measure the concentration of alkalinity-causing ions.

Alkalinity levels can vary depending on the water source and location:

1. Low alkalinity: Less than 10 mg/L as CaCO3 (calcium carbonate)
2. Moderate alkalinity: 10-50 mg/L as CaCO3
3. High alkalinity: Greater than 50 mg/L as CaCO3

- The acceptable limit of alkalinity in drinking water is 200 mg/L as $CaCO_3$ (calcium carbonate), according to IS 10500-2012 [1]. However, in the absence of an alternative source, the permissible limit can be extended to 600 mg/L as $CaCO_3$ [2].

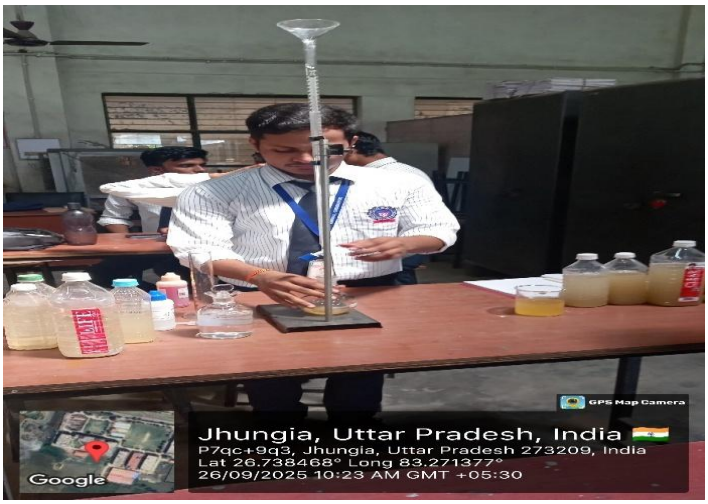


Figure 3.3.1 Alkalinity test

c. Hardness:

- Water hardness refers to the concentration of dissolved minerals, specifically calcium and magnesium, in water. These minerals can cause scaling, which can lead to problems in plumbing, appliances, and industrial equipment.
- Water hardness is typically measured in:
    1. mg/L (milligrams per liter): This unit measures the concentration of calcium and magnesium ions in water.
    2. ppm (parts per million): This unit is equivalent to mg/L.
    3. dGH (degrees of general hardness): This unit measures the concentration of calcium and magnesium ions in water, with 1 dGH equivalent to 17.9 mg/L.
    4. grains per gallon (gpg): This unit measures the concentration of calcium and magnesium ions in water, with 1 gpg equivalent to 17.1 mg/L.

The acceptable limit for water hardness varies depending on the intended use:

1. Drinking water: 200-400 mg/L (11.3-23.4 gpg)
2. Industrial water: 100-500 mg/L (5.7-29.2 gpg)

High water hardness can cause:

1. Scaling: The formation of mineral deposits in pipes and appliances.
2. Corrosion: The degradation of pipes and equipment due to the presence of minerals.
3. Soap scum formation: The formation of soap scum on skin and surfaces.

Figure 3.3.2 Hardness test

d. Turbidity:

- Turbidity refers to the cloudiness or haziness of water due to the presence of suspended particles, such as sediments, clay, silt, and other impurities. Turbidity can affect the appearance, taste, and odor of water, as well as its safety for drinking and other uses.
- Turbidity is typically measured in:
  1. Nephelometric Turbidity Units (NTU): This is the most common unit of measurement for turbidity.
  2. Formazin Turbidity Units (FTU): This unit is similar to NTU but uses a different calibration standard.
  3. WHO (World Health Organization): 5 NTU
  4. USEPA (United States Environmental Protection Agency): 0.3-1 NTU
  5. BIS (Bureau of Indian Standards): 5 NTU

e. Content Chloride:

- Chloride content in water refers to the concentration of chloride ions (Cl-) present in the water. Chloride is a naturally occurring ion that can come from various sources.
- Sources of Chloride-

1. Dissolution of rocks and minerals: Chloride can be released from rocks and mine

2. According to World Health Organization (WHO) guidelines, the desirable limit for chloride in drinking water is **250 mg/L**.

Chloride content in water is typically determined using Mohr's method, a titrimetric procedure. This method involves reacting a water sample with a standard solution of silver nitrate (AgNO3), using potassium chromate (K2CrO4) as an indicator. The endpoint of the titration is marked by the formation of a red-brown precipitate of silver chromate (Ag2CrO4).

3.4. STANDARD VALUE:

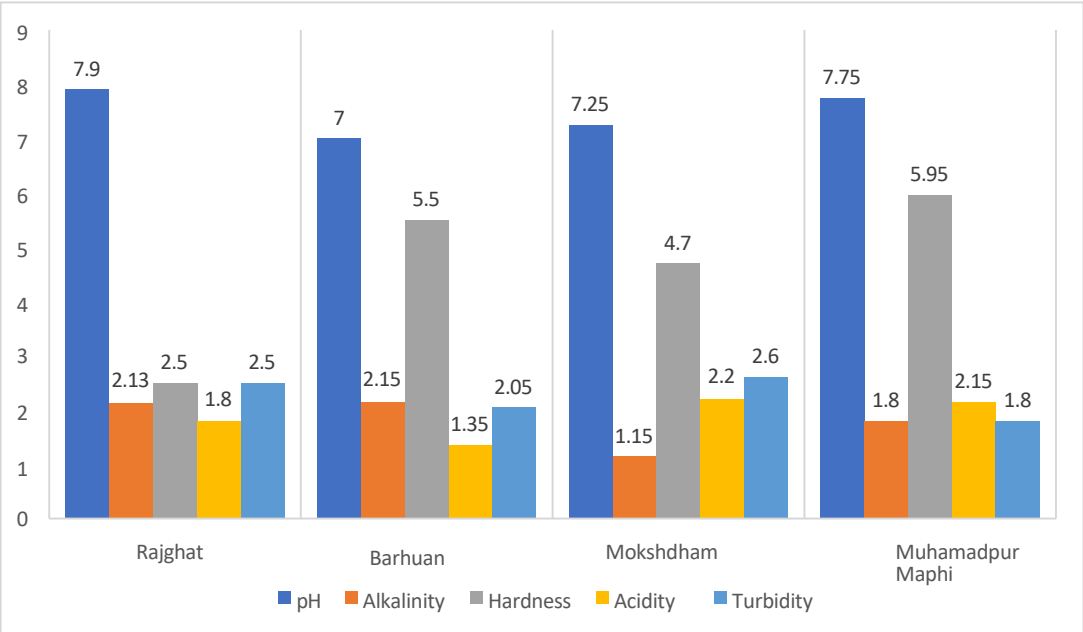
Table 3.4.1: Water Quality Parameters Specified by BIS and WHO

PARAMETER	UNIT	BIS 1991	WHO 2011	ANALYTICAL METHOD
Turbidity	NTU	1	1	Digital Turbidity Meter
Hardness	mg\l	300	600	Titration
pH value	---	6.5-7.5	6.5-7.5	Universal Indicator
Alkalinity	mg\l	200	600	Titration
Chloride content	mg/l	250	250	Titration

4. RESULT & ANALYSIS

Table 4.1: Observation Table

S.No.	Places	pH		Alkalinity (mg/l)		Hardness (mg/l) as $CaCO_3$		Acidity(mg/l)		Turbidity (NTU)	
		S1	S2	S1	S2	S1	S2	S1	S2	S1	S2
1.	Rajghat	8.3	7.5	2.46	1.8	2	3	1.7	1.9	2.6	2.4
2.	Barhuan	6.9	7.1	2.2	2.1	5.3	5.7	1.4	1.3	1.9	2.2
3.	Mokshdham	8	7.9	1.1	1.2	4.3	5.1	2.1	2.3	2.5	2.7
4.	Muhamadpur Maphi	7.5	8	1.7	1.9	5.8	6.1	1.9	2.4	1.4	2.2



Graph 4.2

5. CONCLUSION

The assessment of water quality parameters in the Rapti River revealed significant variations in physical, chemical, and biological characteristics. The results indicate that the river water is polluted with high levels of turbidity, total suspended solids, biochemical oxygen demand, chemical oxygen demand, and total coliform.

The pH, hardness, alkalinity, and chloride content were found to be within the acceptable limits for drinking water, as per the Bureau of Indian Standards. However, the turbidity and bacterial contamination exceeded the permissible limits, posing a risk to human health.

The study highlights the need for effective wastewater management, conservation of natural habitats, and implementation of sustainable agricultural practices to mitigate the pollution of the Rapti River. Regular monitoring and assessment of water quality parameters are essential to ensure the river's water is safe for drinking, irrigation, and other uses.

The study concludes that the Rapti River is facing severe pollution threats due to anthropogenic activities. The high levels of pollutants in the river water pose significant environmental and health risks. Immediate measures are required to control pollution and restore the water quality of the Rapti River. These measures may include:

- Implementation of efficient wastewater treatment systems

REFRENCES

1. Sahni and Varshney; J. Appl. Chem. Sci. Int., vol. 15, no. 2, pp. 1-15, 2024; Article no. JACSI.12328.
2. Md. J. B. Alam, et al. Int. J. Environ. Sci. Tech., 4 (1): 159-167, 2007 ISSN: 1735-1472
3. Km Divya Yadav, Prof. Govind Pandey (Professor)Department of Civil Engineering
4. Madan Mohan Malaviya University of Technology Gorakhpur U.P. India 273010
5. Ayush Kumar Rai; E. Moyo et al. Int. Journal of Engineering Research and Applications
6. ISSN: 2248-9622, Vol. 5, Issue 7, (Part - 4) July 2015, pp.57-61
7. IOSR Journal of Biotechnology and Biochemistry (IOSR-JBB) ISSN: 2455-264X, Volume
8. 3, Issue 4 (Jul. – Aug. 2017), PP 08-15 www.iosrjournals.org
9. M. N. Uddin, M. S. Alam, M. N. Mobin and M. A. Miah; J. Environ. Sci. & Natural Resources,7(1): 253 – 260, 2014 ISSN 1999-7361
10. Pramod Kumar Vishwakarma; International Journal of Engineering Research & Technology (IJERT) ISSN: 2278-0181 Vol. 2 Issue 12, December – 2013
11. IJSRD - International Journal for Scientific Research & Development| Vol. 3, Issue 07, 2015 | ISSN (online): 2321-0613
12. International Research Journal of Engineering and Technology (IRJET) e-ISSN: 2395-0056 Volume: 05 Issue: 09 | Sep 2018 www.irjet.net
13. Prashant Mehta: Archives of Applied Science Research, 2012, 4 (1):497-507
14. Gupta N, Pandey P, Hussain J. Effect of physicochemical and biological parameters on the quality of river water of Narmada, Madhya Pradesh, India" Water Science. 2017
15. International Journal of Computer Sciences and Engineering Vol.7(1), Jan 2019, E- ISSN: 2347-2693
16. Gupta N, Yadav KK, Kumar V, Singh D. Assessment of Physicochemical Properties of Yamuna River in Agra City, International Journal of Chemtech Research. 2015;5(1):528-531 26
17. Rout, Chadetrik, Arun, Lavaniya, Prakash, Divakar. Assessment of Physico-chemical Parameters of River Yamuna at Agra Region of Uttar Pradesh, India. International Research Journal of Environment Sciences. 2015;4:2319-1414.
18. 2020 JETIR June 2020, Volume 7, Issue 6 www.jetir.org (ISSN-2349-5162)